

Program: Diploma in Fire Technology and Safety	
Course Code: 6461B	Course Title: Explosion Theory & Control
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand the chemical and mechanical properties of explosives.
- To learn the fundamentals of detonation, shock waves, and initiation systems.
- To explore the design considerations in explosive applications and their behavior in different environments.
- To learn about the classification, storage, and safe handling practices for explosives.

Course Pre-requisites:

Topic	Course Code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4
Safety in Engineering Industries		Safety in Engineering Industries	4

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	CognitiveLevel
CO1	Understand the chemistry of explosives and their various categories.	14	Understanding
CO2	Learn the mechanisms of sound, shock, and detonation waves, along with initiation systems.	15	Understanding
CO3	Understand scaling, shock waves in different mediums, and explosive device design.	15	Applying
CO4	Learn about the classification, transportation, storage, and safe handling of explosives..	14	Understanding
	SeriesTest	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the chemistry of explosives and their various categories.		
M1.01	Introduction to the chemistry of explosives and chemical reactions.	3	Understanding

M1.02	Explore the different categories of explosives by chemical type.	4	Understanding
M1.03	Understand the mechanics of burning and calculate the state of gas during burning.	4	Understanding
M1.04	Learn about the geometry of explosive grains and interior ballistics.	3	Applying
Contents: Chemical reactions involved in explosive materials. Types of explosives: Propellants, pyrotechnics, and other forms. Mechanics of burning and how explosive grains burn. Basic concepts of interior ballistics.			
CO2	Learn the mechanisms of sound, shock, and detonation waves, along with initiation systems.		
M2.01	Study of sound waves and shock waves in explosive reactions.	3	Understanding
M2.02	Understanding detonation waves and explosive output testing.	4	Understanding
M2.03	Learn the theory of initiation and various types of initiators	4	Understanding
M2.04	Explore non-electric and hot-wire initiation systems.	4	Applying
	SeriesTestI	1	
Contents: Sound and shock waves generated by explosives. Detonation theory and its effects. Initiation theory and different initiator types: Hot-wire, slapper, and bridge wire detonators.			
CO3	Understand scaling, shock waves in different mediums, and explosive device design.		
M3.01	Basics of scaling and geometric similarity in explosive applications.	3	Understanding

M3.02	Learn about accelerating metal with explosives and shock waves in different media.	4	Applying
M3.03	Study of craters formed by explosives and the behavior of conical-shaped charges.	4	Understanding
M3.04	Understand off-the-shelf explosive devices and mechanical/explosive products.	4	Understanding
Contents: Geometric similarity and scaling in explosive design. Behavior of shock waves in air and water. Explosive devices: linear explosive products, mechanical/explosive devices.			
CO4	Learn about the classification, transportation, storage, and safe handling of explosives.		
M4.01	Study the classification of explosives and their transportation.	3	Understanding
M4.02	Learn about the safety practices in storing explosives.	4	Understanding
M4.03	Explore safety protocols in explosive operations, including waste handling.	4	Understanding
M4.04	Understand explosive facility maintenance and licensing requirements.	3	Understanding
	SeriesTest II	1	
Contents: Explosive classification and transportation regulations. Storage of explosives and safety measures. Safe handling, testing, and firing of explosives. Maintenance of explosive facilities and the importance of permits.			

Text/Reference:

T/R	BookTitle/Author
R1	Akhavan, J., "Chemistry of Explosives," Royal Society of Chemistry, 2009.
R2	Kheng, Y., "Explosive Effects and Applications," Springer, 2002.
R3	Mallik, D.V.F., "Explosives and Blasting Techniques," Universities Press, 2005.
R4	Cooper, P.W., & Papalexandris, M.H.V., "Explosives Engineering," Wiley, 1996.